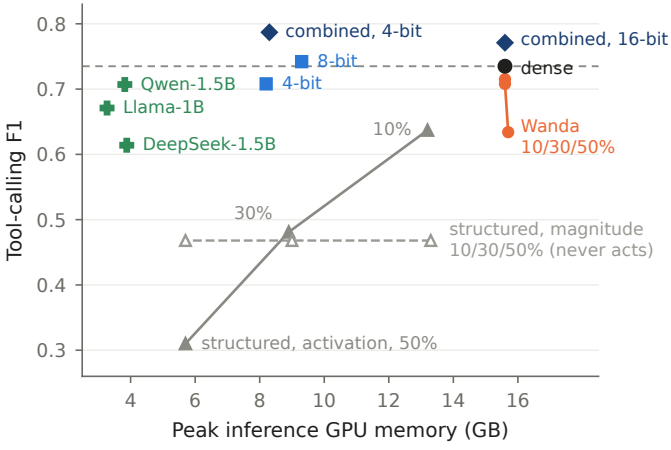
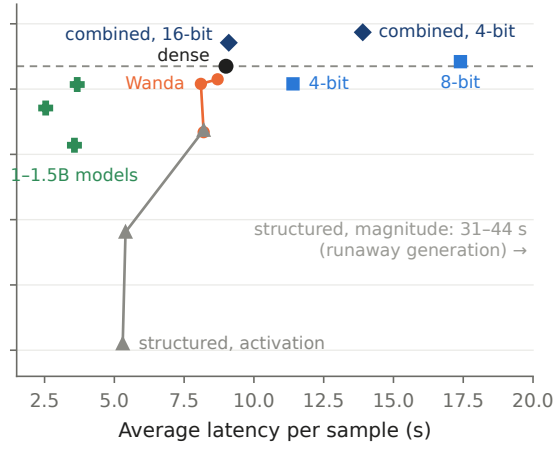

0.8
0.7
0.6
0.5
0.4
0.3
Tool-calling F1
combined, 4-bit
combined, 16-bit
8-bit
dense
Qwen-1.5B
4-bit
Llama-1B
DeepSeek-1.5B
Wanda
10/30/50%
10%
30%
structured, magnitude
10/30/50% (never acts)
structured, activation, 50%
4
6
8
10
12
14
16
Peak inference GPU memory (GB)


combined, 16-bit
dense
combined, 4-bit
Wanda
4-bit
8-bit
1–1.5B models
structured, magnitude: 31–44 s
(runaway generation) →
structured, activation
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Average latency per sample (s)